

Smart Monitoring System For Hybrid Solar Power Plants Using Iot Technology

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ABSTRACT

Solar Power Plants (PLTS) have become one of the most widely used sources of renewable energy across various sectors, including both commercial industries and households. A hybrid solar power system combines solar energy with other sources, such as generators or the electrical grid. The urgency of this research lies in the development of an Internet of Things (IoT)-based monitoring system for hybrid solar power plants, which can analyze data in real time, facilitate problem detection, optimize performance, and design preventive maintenance, as well as serve as an alternative source of electrical energy. The objective of this study is to develop a monitoring system for hybrid solar power plants using IoT. This technology is essential for enhancing the efficiency and sustainability of hybrid solar power generation. The method used in this research is the experimental method. Solar panels function as energy collectors that convert sunlight into electrical energy. The solar charge controller regulates the amount of electrical energy from the panels used to charge the battery. The voltage, current, and power generated by the plant are detected by two sensors: a DC voltage sensor and an ACS712 current sensor. The energy stored in the battery is used to power lamps. IoT offers an efficient solution for integrating components of a solar power plant, such as solar panels, inverters, batteries, and others. A hybrid solar power monitoring system can ensure maximum energy utilization and reduce long-term operational costs.

INTRODUCTION

Solar Power Plants (PLTS) have become one of the most widely used sources of renewable energy in various sectors, including commercial industries and households. A hybrid solar power system combines solar energy with other sources, such as generators or the electrical grid. Based on the above background, the research problem to be addressed is how to develop a hybrid solar power plant monitoring system using IoT. The urgency of this research lies in developing an IoT-based hybrid solar power monitoring system that can analyze data in real time, facilitate problem detection, optimize performance, design preventive maintenance, and serve as an alternative source of electrical energy.



Figure 1. Solar panel

This renewable energy-based power plant is one of the recommended solutions for electricity in remote rural areas, where sunlight is abundant and fuel is relatively expensive and difficult to obtain. The photovoltaic system converts the electromagnetic energy from sunlight.

This renewable energy-based power plant is one of the recommended solutions for providing electricity in remote rural areas, where sunlight is abundant and fuel is scarce and relatively expensive. There are several types of solar power plants (PLTS), including systems connected to the PLN power grid (on-grid). They are commonly applied



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in remote areas. Off-grid systems work by having solar panels generate electricity, which is stored in batteries and used as needed.

The photovoltaic module is the main component of a solar power plant (PLTS), and its quality is crucial to maintaining system operation. The frame is made of anodized aluminum to prevent corrosion. The frame serves to ensure the rigidity of the panel, which consists of protective glass that shields it from environmental factors and maintains the panel's structural strength. The frame installation is carried out at the final stage of the manufacturing process, including the protective glass and encapsulation or lamination. A layer is applied between the photovoltaic cells and the protective glass to prevent damage to the cells, using ethylene-vinyl acetate (EVA) material. The photovoltaic cells are made of semiconductor materials that capture sunlight and convert it into electricity.

LITERATURE REVIEW

In study This renewable energy-based power plant is one of the recommended solutions for providing electricity in remote rural areas, where sunlight is abundant and fuel is scarce and relatively expensive. (Abed, 2018), it is stated that previous research on monitoring still had shortcomings because Bluetooth was used for system communication, thus limiting the monitoring range. Previous studies on the monitoring of solar cell and wind power plants using IoT are found in (Madadi, 2021)(Mucahit Cihan, 2019). Web-based solar power plant monitoring using the open-source IoT platform ThingSpeak and Arduino was conducted in (Kandimalla & Kishore, 2017). The next study discusses the design and implementation of a low-cost smart IoT solution for photovoltaic station energy monitoring (Cheddadi et al., 2020). Study (Priharti et al., 2019) addresses the design of a real-time monitoring application to assess the performance of photovoltaic systems. IoT services are designed to enable remote control and obtain sensor data from Arduino or microcontroller devices using a dedicated application, namely the Blynk App (Priharti et al., 2019). Further research involves monitoring system development using prototypes (Artiyasa et al., 2021). Real-time monitoring of solar panel parameters based on Arduino is discussed in (Rus-Casas et al., 2020). Study (Fachri et al., 2015) covers a monitoring system for temperature, humidity, and irrigation control in hydroponic plants using Blynk for Android. Research on IoT-based motorcycle monitoring is presented in , along with several studies on real-time monitoring of Arduino-based solar home systems (Ramady et al., 2021). In (Zhong, 2024), an IoT-based monitoring system for solar thermal stations is discussed. Study (Rahman et al., 2018) examines a modern global monitoring system for renewable energy power plants. The intelligent solar power system unit monitors light intensity, voltage, and current between the battery and the DC-to-AC converter, as well as the voltage generated by the solar panel(Herman & Surantha, 2020). The values of temperature, pH, and water level in the water tank, along with light intensity, current, and voltage, can then be transmitted to a server via wireless communication(Siregar et al., 2016).

The above studies share similarities with the present research in terms of the monitoring theme. However, the methods and research objects differ, with our approach being supported by IoT(Ho et al., 2019). For data processing from these objects, data analysis will be supported by the causal method combined with an artificial neural network method as part of an artificial intelligence system.

Based on the state of the art described above, the novelty of this study lies in the monitoring technology for hybrid solar power plants using the Internet of Things. Data analysis is supported by the neural network method and optimized using Particle Swarm Optimization. Figure 2 illustrates the gap analysis to highlight the novelty of this research.

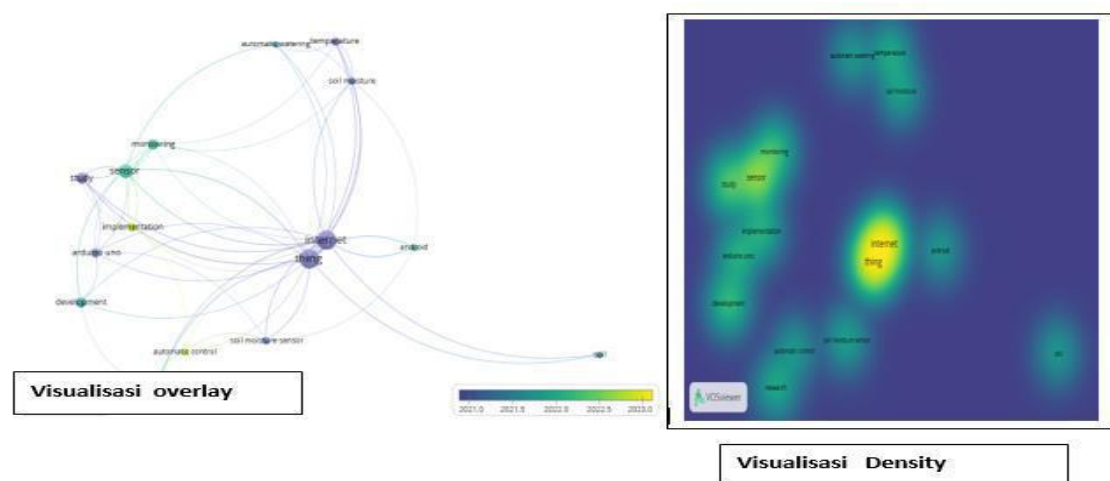


Figure 2. Gap Analysis

In Figure 1, the overlay visualization shows that IoT-based research is currently trending, thus offering the potential to produce novelty compared to other related studies(Lestari et al., 2021).

METHOD

In this study, the experimental method is used. We designed an IoT-based hybrid solar power plant monitoring system. The solar panel functions as an energy collector that converts sunlight into electrical energy. The solar charge controller regulates the amount of electrical energy from the panel used to charge the battery. The voltage, current, and power generated by the plant are detected by two sensors: a DC voltage sensor and an ACS712 current sensor. The energy stored in the battery is used to power lamps. The data produced—consisting of voltage, current, and power from solar radiation and wind captured by the solar panel and wind turbine—can be monitored through the Blynk application. This provides an efficient solution for integrating components of a solar power plant, such as solar panels, inverters, batteries, and others, with IoT.

The problem-solving approach involves designing a device developed for a hybrid solar power plant monitoring system using IoT. The system consists of a photovoltaic module, solar controller, battery, and several other supporting components. The solar panel functions as an energy collector that converts sunlight into electrical energy. The solar charge controller regulates the amount of electrical energy from the panel used to charge the battery. The voltage, current, and power generated by the plant are detected by two sensors: a DC voltage sensor and an ACS712 current sensor. The energy stored in the battery is used to power lamps.(Herman & Surantha, 2020)

IoT offers an efficient solution for integrating components of The sensor data will be checked to determine if it falls within the specified range.a solar power plant, such as solar panels, inverters, batteries, and others. The data produced—comprising voltage, current, and power from solar radiation and wind captured by the solar panel and wind turbine—can be monitored through the Blynk application. A hybrid solar power monitoring system can ensure maximum energy utilization and reduce long-term operational costs.

The monitoring system flowchart is shown in Figure 3 below.

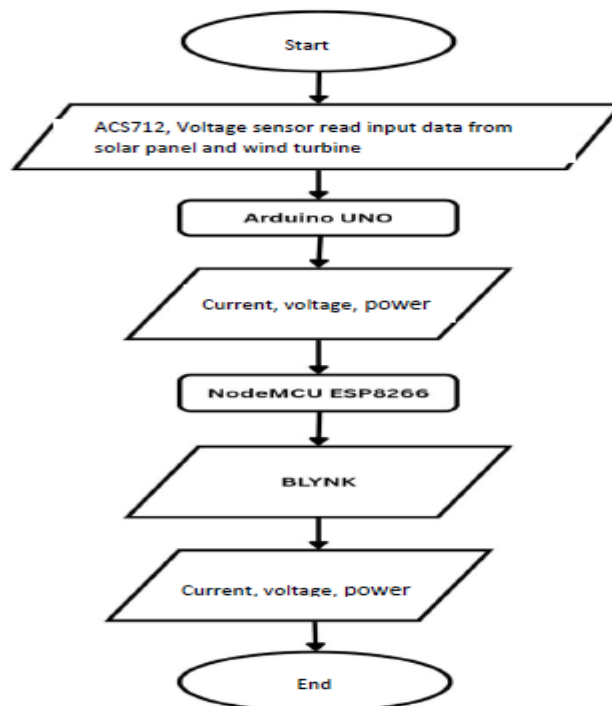


Figure 3. Monitoring System Flowchart

RESULT

The performance of a solar panel installed under specific environmental conditions can be determined by directly monitoring its output parameters such as voltage, current, and power. From this monitoring, it is possible to assess whether the installation of the solar panel is appropriate and whether it produces the expected output power. The current method of monitoring solar panels only collects output parameter data in the form of text files with a specific format.

The solar panel output parameter monitoring technique introduced uses an Arduino Atmega 328P microprocessor, PLX-DAQ data acquisition that can be directly integrated with Microsoft Office Excel, a current



sensor, a voltage sensor, a power supply, and a computer device. The equipment structure of the monitoring system used can be seen in Figure 4.

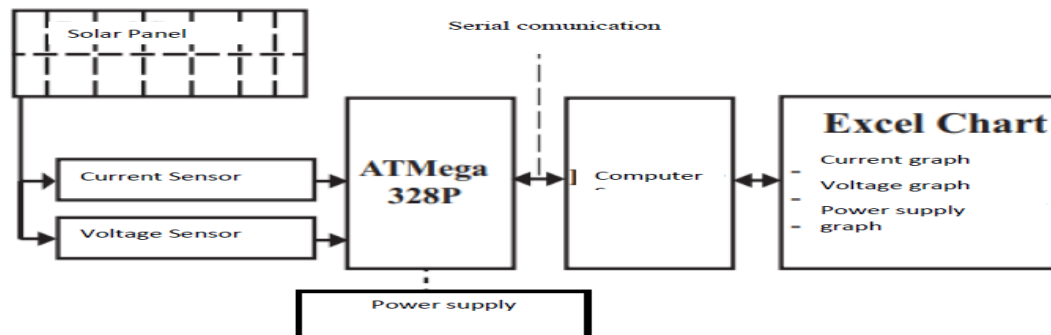


Figure 4. Schematic of solar panel parameter monitoring

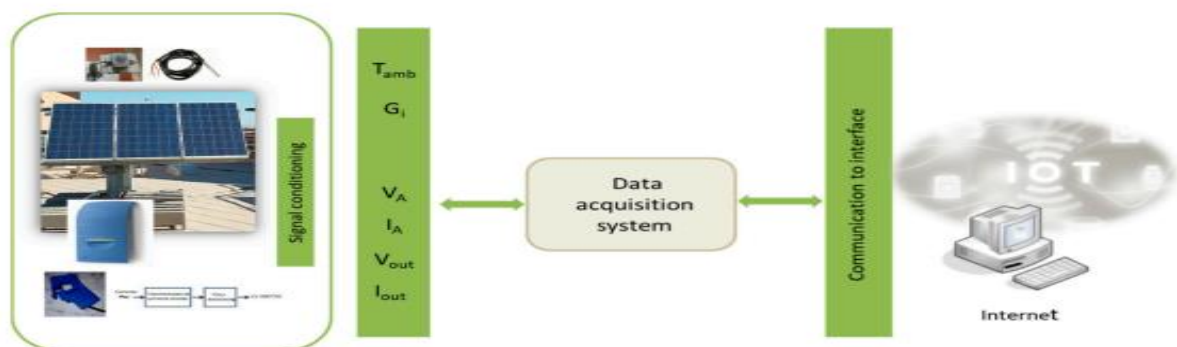


Figure 5. The block diagram of the prototype for monitoring a photovoltaic self-consumption system

The sensor block includes weather and electrical sensors with built-in electronics that adjust the output signals to the correct levels. The equipment for monitoring the output parameters of the solar panel is shown in Figure 4. The microcontroller receives the solar panel parameter readings from the sensor module and sends them to the computer via serial communication using an RS232 connection. The sensor reading data is presented in Figure 6.



Figure 6. PLX-DAQ program for interface communication.

The computer used must be very fast to support real-time data acquisition. Since the program is used for interface communication, the performance of the solar panel can be directly monitored with voltage and current samples taken approximately every 1 minute during the research. However, displaying the graph will increase processing time.

The power electronic component in a solar power plant (PLTS) regulates battery charging using photovoltaic modules to achieve optimal performance. This device can control the charging voltage and current of the battery. To reach higher charging currents, multiple solar charge controllers can be connected in parallel to combine the power from photovoltaic modules.

DISCUSSION

The solar panel functions as an energy collector, which is then converted into electrical energy. A controller regulates the amount of electrical energy from the panel used to store power in the battery. The power stored in the battery is used to operate the lights.

A hybrid solar power plant (PLTS Hybrid) is a photovoltaic system that, as the name suggests, combines solar energy with other power sources such as the utility grid (PLN), generators, or other energy sources like wind turbines. This system can be equipped with batteries as backup, allowing it to supply power at night or during power outages.

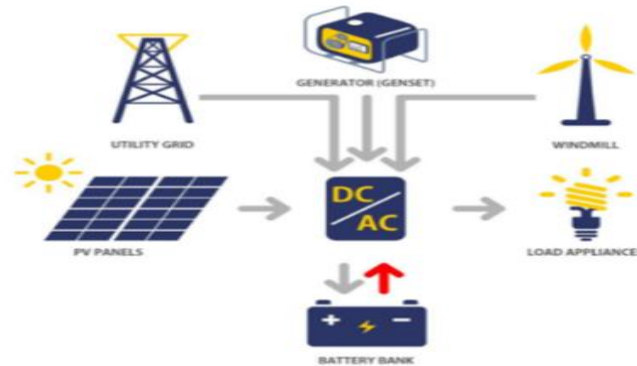


Figure 7. Hybrid Solar Power System (PLTS Hybrid System)

The explanation of Figure 7 is as follows. A hybrid solar power system integrates photovoltaic panels with other energy sources such as diesel generators, wind turbines, or the utility grid to ensure a stable and reliable electricity supply. Solar energy is prioritized during the day, while batteries or backup sources cover shortages at night or during low sunlight. This system offers continuous power, improved fuel efficiency, flexibility for remote or grid-connected areas, and reduced emissions. However, it requires higher initial investment and more complex maintenance compared to single-source systems.

After installing the system, the solar panel system is tested by measuring the output voltage, current, and power generated. The calibration results of the voltage sensor are presented in Table 1.

Table 1 . Voltage sensor calibration results.

No.	Multimeter Volt	Mikrokontroler Volt	Galat (%)
1.	0	0	0
2.	2,04	2,03	0,49
3.	4,07	4,11	0,98
4.	12,3	12,6	0,25

The voltage monitoring graph is shown in Figure 8. A real-time measurement and monitoring system is used to assess the performance of solar panels. The configuration is assumed to consist of voltage and current sensors, enabling the monitoring of the power absorbed by the solar panel load. An Arduino integrated with these sensors allows data acquisition while simultaneously monitoring the solar panel's performance.

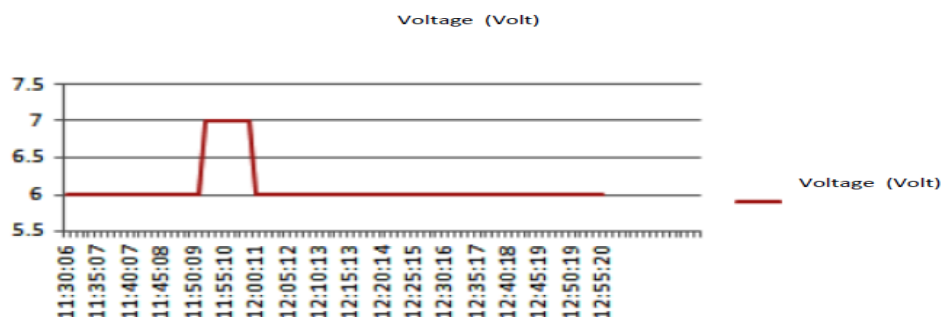


Figure 8. The voltage monitoring graph

The data measured by the sensors can be transmitted via RS232 and analyzed to produce performance graphs of the solar panel. The solar panel has a capacity of 2×50 Wp connected in parallel. It is loaded with two 50-watt lamps connected in series. The real-time monitoring results of the solar panel can be viewed in MS Excel, recorded directly through the PLX-DAQ interface program.

CONCLUSION

IoT technology is an efficient solution for simplifying the monitoring and management of this system. Devices are connected to various components of the solar power system, such as solar panels, inverters, batteries, and others. All data is analyzed in real time, making it easier to detect problems, optimize performance, and plan preventive maintenance.. A hybrid solar power monitoring system ensures maximum energy utilization, speeds up problem detection and repair, and reduces long-term operational costs. This makes IoT technology a vital tool for improving the efficiency and sustainability of hybrid solar power plants.

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